

Apache OpenOffice

Version 4.1

Calc Guide

AOO Documentation Team

Chapter 3

Creating Charts and Graphs

Presenting information visually

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Note for Mac users

Some keystrokes and menu items differ on a Mac from those used in Windows and Linux. The table below gives some common substitutions for the instructions in this chapter. For a more detailed list, see the application Help.

<i>Windows/Linux</i>	<i>Mac equivalent</i>	<i>Effect</i>
Tools > Options menu selection	OpenOffice > Preferences	Access setup options
<i>Right-click</i>	<i>Control+click</i>	Open context menu
<i>Ctrl</i> (Control)	⌘ (Command)	Used with other keys
<i>F5</i>	<i>Shift+⌘+F5</i>	Open the Navigator
<i>F11</i>	<i>⌘+T</i>	Open Styles & Formatting window

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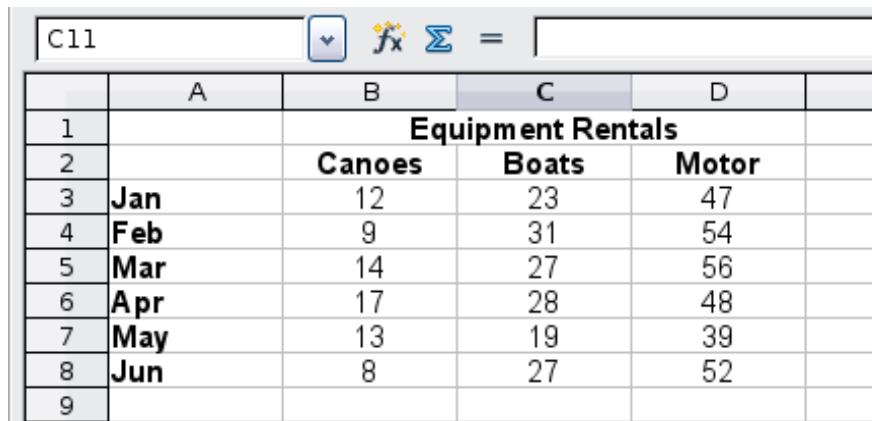
Introduction

Charts and graphs can be powerful ways to inform users. OpenOffice Calc offers a variety of chart and graph formats for your data.

With Calc, you can customize charts and graphs considerably, allowing you to present information in the most clear and informative way possible.

Creating a Chart

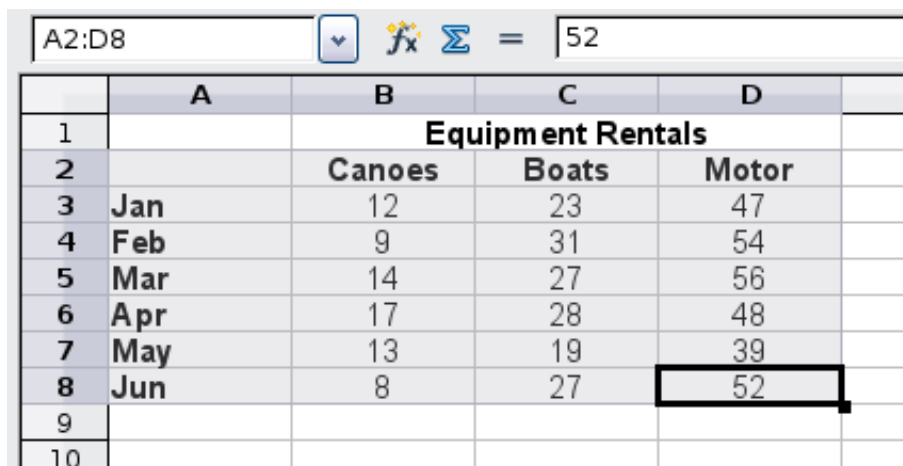
To demonstrate the process of making charts and graphs in Calc, we will use the small table of data in Figure 1.



	A	B	C	D
1		Equipment Rentals		
2		Canoes	Boats	Motor
3	Jan	12	23	47
4	Feb	9	31	54
5	Mar	14	27	56
6	Apr	17	28	48
7	May	13	19	39
8	Jun	8	27	52
9				

Figure 1: Table of data for charting examples

To create a chart, first highlight (select) the data to be included in the chart. The selection does not need to be in a single block, as shown in Figure 2; you can also choose individual cells or groups of cells (columns or rows). See Chapter 1 (Introducing Calc) for more information about selecting cells and ranges of cells.



	A	B	C	D
1		Equipment Rentals		
2		Canoes	Boats	Motor
3	Jan	12	23	47
4	Feb	9	31	54
5	Mar	14	27	56
6	Apr	17	28	48
7	May	13	19	39
8	Jun	8	27	52
9				
10				

Figure 2: Selecting data for plotting

Next, open the Chart Wizard dialog using one of two methods.

- Choose **Insert > Chart** from the menu bar.
- Or, click the **Chart** icon on the main toolbar.

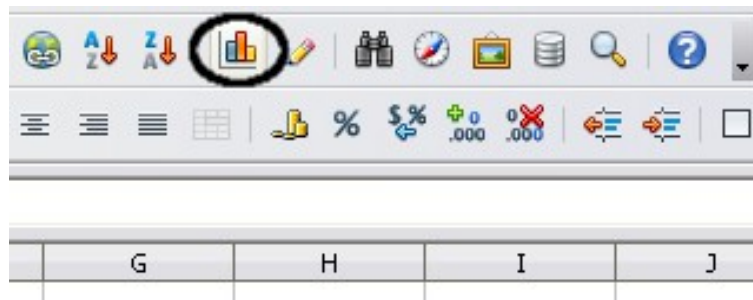


Figure 3: Insert chart from main toolbar

Either method inserts a sample chart on the worksheet, opens the Formatting toolbar, and opens the Chart Wizard, as shown in Figure 4.

Tip

Before choosing the Chart Wizard, place the cursor anywhere in the area of the data. The Chart Wizard will then do a fairly good job of guessing the range of the data. If the Chart Wizard struggles to select the data you want, select the cells manually before starting the Chart Wizard.

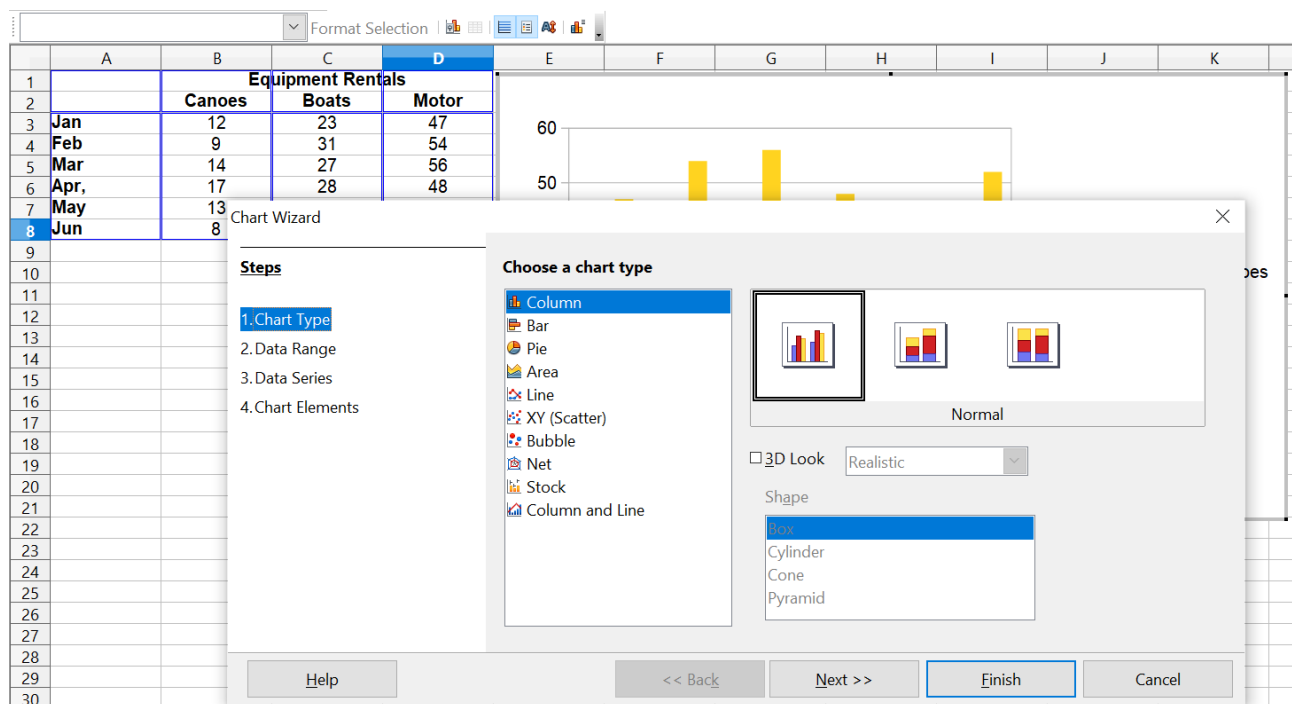


Figure 4: Chart Wizard, Step 1—Choose a chart type

Choosing a Chart Type

The Chart Wizard includes a sample chart with your data. This sample chart updates to reflect your changes in the Chart Wizard.

The Chart Wizard has three main parts:

- a list of steps involved in setting up the chart
- a list of chart types
- the options for each chart type.

At any point, when using Chart Wizard, you can go back to a previous step and change selections.

Calc offers a choice of 10 basic chart types with a few options for each. The options vary according to the type of chart you pick.

The first tier of choice is for two-dimensional (2D) charts. Only those types suitable for 3D (Column, Bar, Pie, and Area) give you the option to select a 3D aspect.

On the *Choose a chart type* page (Figure 4), select a type by clicking its icon. The preview updates every time you select a different type of chart and provides a good idea of what the finished chart will look like.

The current selection is highlighted (shown with a surrounding box) on the *Choose a chart type* page. The chart's name is displayed just below the icons. For now, we will stick to the Column chart and click on **Next** again.

Changing Data Ranges and Axes Labels

In Step 2, Data Range, you can manually correct any mistakes made in selecting the data.

On this page, you can also change how you plot data by using rows—rather than columns—as data series. The Chart Wizard attempts to set this and the following setups automatically, according to the layout of the data. (This often works, but you may need to correct the default settings.)

Lastly, you can choose to use the first row, first column, or both as labels on the chart's axes.

You can confirm what you have done so far by clicking the **Finish** button or clicking **Next** to change some details of the chart.

Now we'll click **Next** to investigate more changes available to our chart from other areas of the Wizard.

Steps

1. Chart Type
2. Data Range
3. Data Series
4. Chart Elements

Choose a data range

Data range
\$Sheet1.\$A\$2:\$D\$8

☐ Data series in rows
☒ Data series in columns

☒ First row as label
☒ First column as label

Help << Back Next >> Finish Cancel

Figure 5: Changing data ranges and axes labels

Selecting Data series

Steps

1. Chart Type
2. Data Range
3. Data Series
4. Chart Elements

Customize data ranges for individual data series

Data series	Data ranges
Canoes	Name: \$Sheet1.\$B\$2 Y-Values: \$Sheet1.\$B\$3:\$B\$8
Boats	
Motors	

Range for Name
\$Sheet1.\$B\$2

Categories
\$Sheet1.\$A\$3:\$A\$8

Add Remove

Help << Back Next >> Finish Cancel

Figure 6: Amending data series and ranges

On the Data Series page, you can fine-tune the data you'd like to include in the chart. Perhaps you've decided that you do not want to include data for canoes. If so, highlight *Canoes* in the **Data series** box and click **Remove**. Each named data series has its ranges and individual Y-values listed. This is useful if you have very specific requirements for data in your chart, as you can include or leave out these ranges. For example, you could use the **Add** button to include y values from a distinct area, such as H16:H22, by setting the *Range for Name* to H16 and the *Y-Values* to H17:H22. This would work as long as the categories of the data in H17:H22 matched the order in A3:A8, where the rest of the data are.

Tip

You can click the Shrink button  next to the *Range for Name* box to work on the spreadsheet itself. This is handy if your data ranges are larger than those in our samples and the Chart Wizard is in the way.

Click **Next** to deal with titles, legend, and grids.

Adding or Changing Titles, Legend, and Grids

On the Chart Elements page (Figure 7), you can give your chart a title and, if desired, a subtitle. Use a title that draws the viewers' attention to the purpose of the chart: what you want them to see. For example, a better title for this chart might be *The Performance of Motor and Other Rental Boats*.

It is almost always beneficial to have labels for the x-axis or the y-axis. That allows you to orient viewers by giving them an idea of the meaning, orientation, and proportion of your data. For example, if we put Thousands in the y-axis label of our graph, it changes the scope of the chart entirely. For ease of estimating data, you can also display the x- or y-axis grids by selecting the *Display grids* options.

You can omit a chart's legend or include it and place it to the left, right, top, or bottom.

To confirm your selections and complete the chart, click **Finish**.

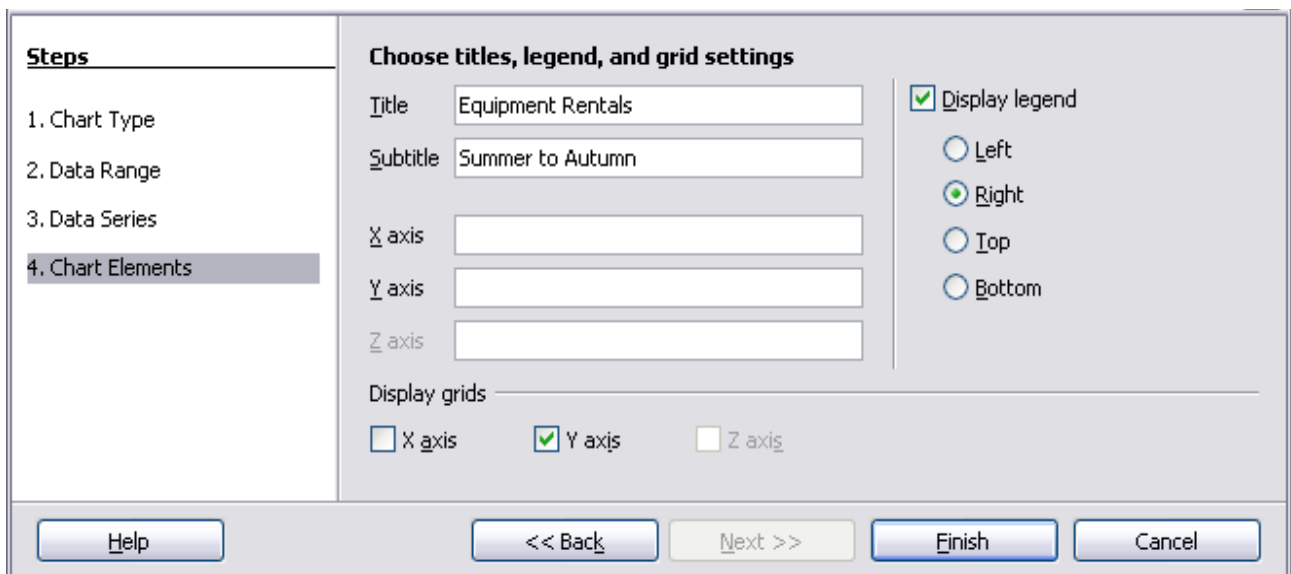


Figure 7: Titles, legend and grids

Editing Charts

After you have created a chart, you may find things you would like to change. Calc provides tools for changing the chart type, chart elements, data ranges, fonts, colors, and many other options through the **Insert** and **Format** menus, the right-click (context) menu, and the Chart toolbar.

Changing the Chart Type

You can change the chart type at any time. To do so:

- 1) First, select the chart by double-clicking on it. A gray border should now surround the chart.
- 2) Then do one of the following:
 - Choose **Format > Chart Type** from the menu bar.
 - Click the chart type icon  on the Formatting toolbar.
 - Right-click on the chart and choose **Chart Type**.

In each case, a dialog similar to the one in Figure 4 opens. See page 6 for more information.

Adding or Removing Chart Elements

Figures 8 and 9 show the elements of 2D and 3D charts.

The default 2D chart includes only two of those elements:

- *Chart wall* contains the graphic of the chart displaying the data.
- *Chart area* is the area surrounding the chart graphic. The (optional) chart title and the legend (key) are in the chart area.

The default 3D chart also has the *chart floor*, which is unavailable in 2D charts.

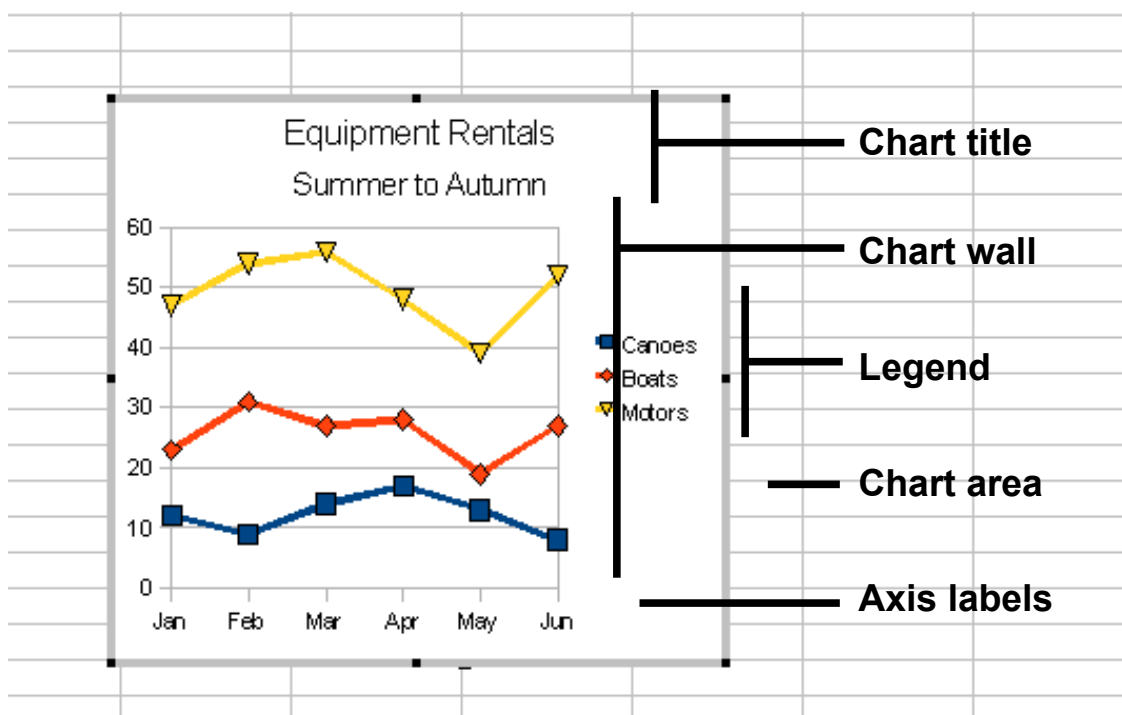


Figure 8: Elements of 2D chart

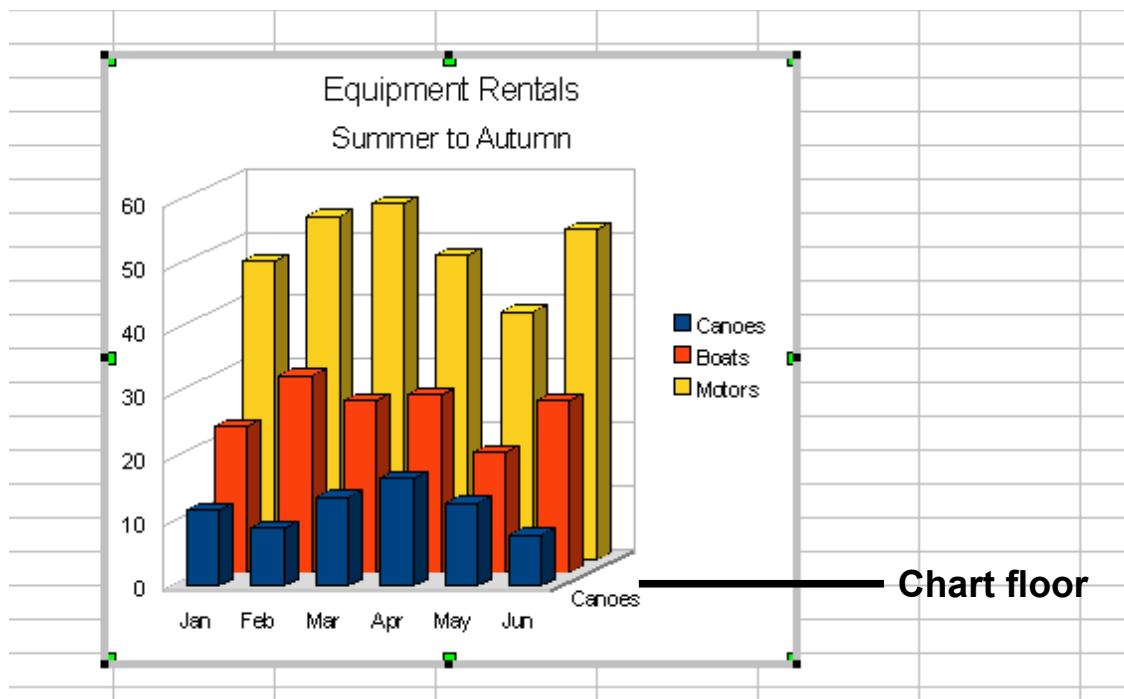


Figure 9: Elements of 3D chart

You can add other elements using the commands on the **Insert** menu. The various choices open dialogs in which you can specify details.

First, select the chart so the green sizing handles are visible. This is done with a single click on the chart.

The dialogs for Titles, Legend, Axes, and Grids are self-explanatory. The others are a bit more complicated, so we'll look at them now.

Data Labels

Data labels supply information about each data point on the chart. They can be very useful for presenting detailed information, but you need to be careful to avoid creating a chart that is too cluttered to read.

Select the graph with a double click and choose **Insert > Data Labels**. The options are as follows.

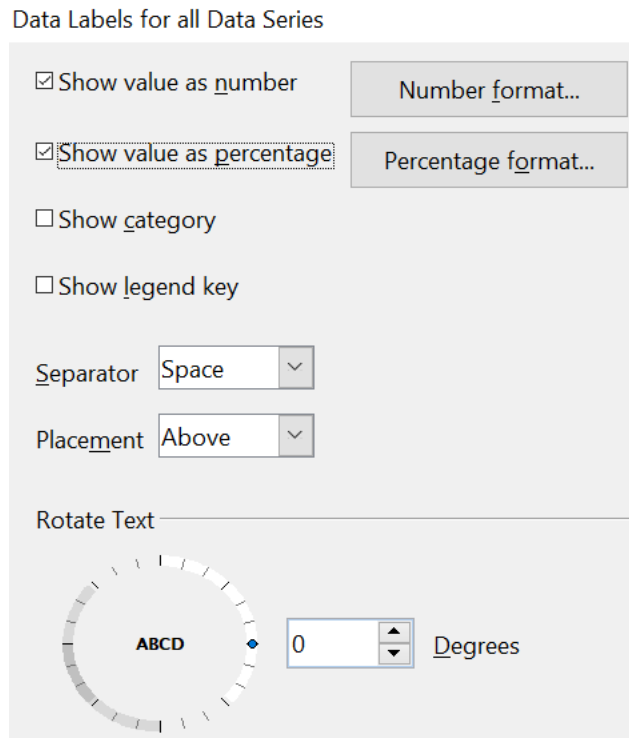


Figure 10: Data Labels dialog

Show value as number

Displays the numeric values of the data points. When selected, this option activates the **Number format...** button.

Number format...

Opens the Number Format dialog, where you can select the number format. This dialog is very similar to the one for formatting numbers in cells, described in Chapter 2 (Entering, Editing, and Formatting Data).

Show value as percentage

Displays the percentage value of the data points in each column. When selected, this option activates the **Percentage format...** button.

Percentage format...

Opens the Number Format dialog, where you can select the percentage format.

Show category

Shows the data point text labels.

Show legend key

Displays the legend icons next to each data point label.

Separator

Selects the separator between multiple text strings for the same object.

Placement

Selects the placement of data labels relative to the objects.

Figure 23 on page 27 shows examples of values as text (neither *Show value as number* nor *Show value as percentage* selected) and values as percentages, as well as when data values are used as substitutes for legends or in conjunction with them.

Trend Lines

When you have a scattered grouping of points in a graph, you may want to show any relationship among the points. You can use a trend line to do so. Calc has a good selection of regression types for trend lines. These include:

- linear
- logarithm
- exponential
- power

Choose the type that comes closest to passing through all of the points or that represents an expected relationship.

Tip

Logarithmic regression predicts y-values that lie outside points already plotted. Exponential regression is used to model situations in which growth begins slowly and then accelerates rapidly or in which decay begins rapidly and then slows down. Power Regression involves a response variable proportional to a control variable that's been raised to a specific power.

Tip

Regression can be viewed as a functional relationship between two or more variables. That relationship is often determined from data, and used to predict values of one entity when given values of others. One example? The statement 'the regression of y on x is linear' means that the value of y depends on the value of x.

To insert trend lines for all data series, double-click the chart to enter edit mode. Choose **Insert > Trend Lines**, then select the type of trend line from *None*, *Linear*, *Logarithmic*, *Exponential*, or *Power*. You can also choose whether to show the equation for the trend line and the coefficient of determination (R^2).

To insert a trend line for a single data series, first select the data series in the chart, then right-click and choose **Insert > Trend Line** from the context menu. The dialog for a single trend line is similar to the one below but has a second tab (Line), where you can choose the line's attributes (style, color, width, and transparency).

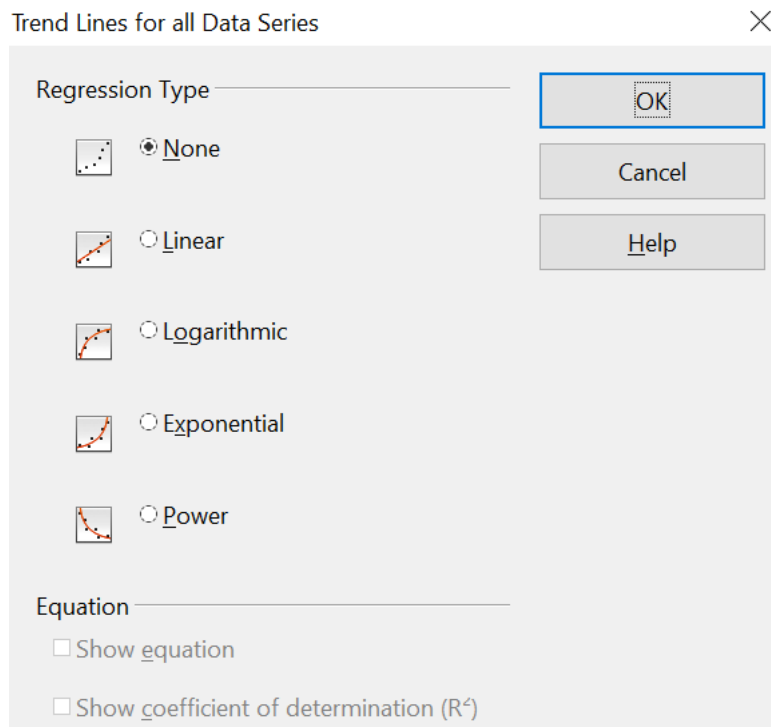


Figure 11: Trend Lines dialog

To delete a single trend line or mean value line, click the line, then press the *Del* key.

To delete all trend lines, choose **Insert > Trend Lines**, then select **None**.

A trend line is shown in the legend automatically.

If you insert a trend line on a chart type that uses categories, such as Line or Column, then the numbers 1, 2, 3, ... are used as x-values to calculate the trend line. Such trend lines can be useful visual aids, but the equation of the trend line may not be meaningful.

The trend line has the same color as the corresponding data series. To change the line properties, right-click the trend line and choose **Format Trend Line**. This opens the *Line* tab of the Trend Lines dialog.

To show the trend line equation, select the trend line in the chart, right-click to open the context menu, and choose **Insert Trend Line Equation**.

When the chart is in edit mode, OpenOffice gives you the equation of the trend line and the correlation coefficient. Click on the trend line to see the information in the status bar. To show the equation and the correlation coefficient, select the line and choose **Insert R² and Trend Line Equation**.

Tip

We can define a correlation coefficient as a value that describes the degree of correlation between two sets of data or between two random variables.

For more details on the regression equations, see the topic *Trend lines in charts* in the Help.

Mean Value Lines

If you select the Mean Value Lines dialog from the Insert menu, Calc calculates the average of each selected data series and places a colored line at the correct level in the chart.

Y Error Bars

If you are presenting data that has a known possibility of error, such as social surveys using a particular sampling method, or you want to show the measuring accuracy of the tool you used, you may wish to show error bars on the chart. Select the chart and choose **Insert > Y Error Bars**.

Several options are provided in the Error Bars dialog. You can only choose one option at a time. You can also choose whether the error indicator shows both positive and negative errors or only positive or only negative.

Error Bars for Data Series 'Motor'

Y Error Bars Line

Error Category	Error Indicator
☐ Constant Value	☒ Positive and Negative
☐ Percentage	☐ Positive
☒ Standard Deviation	☐ Negative
☐ Cell Range	

Parameters

Positive (+)

Negative (-)

☒ Same value for both

Figure 12: Specifying the parameters of error bars

- **Constant Value** – you can have separate positive and negative values.
- **Percentage** – choose the error as a percentage of the data points.
- In the drop-down list:
 - **Standard Error** – calculates the error based on the numerical data you provide in the chart
 - **Variance** – shows error calculated on the size of the biggest and smallest data points
 - **Standard Deviation** – shows error calculated on standard deviation
 - **Error Margin** – you designate the error
- **Cell Range** – calculates the error based on cell ranges you select. The Parameters section at the bottom of the dialog changes to allow selection of the cell ranges.

Tip

Standard deviation measures how dispersed data is in relation to its mean. Low or small standard deviation indicates data clustered tightly around the mean. High or large standard deviation indicates dispersion that is more spread out.

Formatting Charts

The Format menu has many options for fine-tuning the appearance of your charts.

Double-click the chart so it is enclosed by a gray border indicating edit mode; select the chart element you want to format. Choose **Format** from the menu bar, or right-click to display a context menu relevant to the selected element. The formatting choices are as follows.

Format Selection

Opens a dialog in which you can specify the area fill, borders, transparency, characters, font effects, and other attributes of the selected element of the chart (see page 20).

Position and Size

Opens a dialog (see page 24).

Arrangement

Provides two choices: **Bring Forward** and **Send Backward**, of which only one may be active for some items. Use these choices to arrange overlapping data series.

Title

Formats the titles of the chart and its axes.

Legend

Formats the location, borders, background, and type of the legend.

Axis

Formats the lines that create the chart as well as the font of the text that appears on both the X and Y axes.

Grid

Formats the lines that create a grid for the chart.

Chart Wall, Chart Floor, Chart Area

These functions are described in the following sections.

Chart Type

Changes what kind of chart is displayed and whether it is two- or three-dimensional.

Data Ranges

Explained on page 6 (Figure 5 and Figure 6).

3D View

Formats 3D charts (see page 17).

Note

Chart Floor and **3D View** are only available for a 3D chart. These options are unavailable (grayed out) if a 2D chart is selected.

In most cases, you need to select the exact element you want to format. Sometimes this can be tricky to do with the mouse, especially if the chart has many elements, some of which may be small or overlapping. If you have Tooltips turned on (in **Tools > Options > OpenOffice > General > Help**, select **Tips**), then as you move the mouse over each element, its name appears in the Tooltip. Once you have selected one element, you can press *Tab* to move through the other elements until you find the one you want. The name of the selected element appears in the Status Bar.

Moving Chart Elements

You can move or resize individual elements of a chart independent of other such elements. For example, you can move a chart's legend to a different place. Pie charts allow moving individual wedges of the pie (in addition to the choice of “exploding” the entire pie).

- 1) Double-click the chart so that it is enclosed by a gray border.
- 2) Double-click any of the elements—the title, the legend, or the chart graphic. Click and drag to move the element. If the element is already selected, move the pointer over the element to get the move icon (small hand), click, drag, and then move the element.
- 3) Release the mouse button when the element is in the desired position.

Note

If your chart graphic is 3D, round red handles appear which control the three-dimensional angle of the graphic. You cannot resize or reposition the graphic while the round red handles are showing. With the round red handles showing, *Shift+Click* to get the green resizing handles. You can now resize and reposition your 3D chart graphic. See the following tip.

Tip

You can resize the chart graphic using its green resizing handles (*Shift+Click*, then drag a corner handle to maintain the proportions). However, you cannot resize the title or the key.

Changing the Chart Area Background

The chart area is the area surrounding the chart graphic, including the (optional) main title and key.

- 1) Double-click the chart so that it is enclosed by a gray border.
- 2) Choose **Format > Chart Area** or right-click on the chart area and choose **Format Chart Area**.
- 3) On the Chart Area dialog, choose the desired format settings.

On the *Area* tab, you can change the color, or choose a hatch pattern, bitmap or preset gradients. Click on the drop-down box to see the options. Patterns are probably more useful than color if you have to print out your chart in black and white.

You can also use the *Transparency* tab to change the area's transparency. If you used a preset gradient from the *Area* tab, you can see the different parameters of which it is composed.

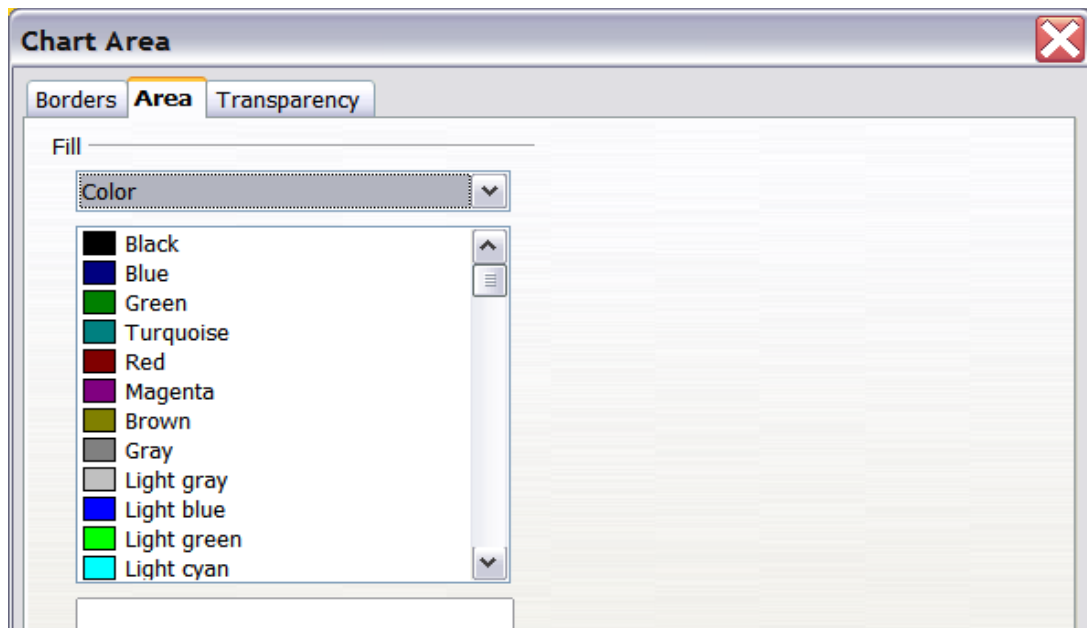


Figure 13: Chart Area dialog

Changing the Chart Graphic Background

The chart wall is the area that contains the chart graphic.

- 1) Double-click the chart so that it is enclosed by a gray border.
- 2) Choose **Format > Chart Wall**. The Chart Wall dialog has the same formatting options as described in "Changing the Chart Area Background" above.
- 3) Choose your settings and click **OK**.

Changing Colors

If you need a different color scheme from the default for the charts in all your documents, go to **Tools > Options > Charts > Default Colors**, which has a much wider range of colors to choose from. Changes made in this dialog affect the default chart colors for any chart you make in the future.

Formatting 3D Charts

Use **Format > 3D View** to fine-tune 3D charts. The 3D View dialog has three tabs where you can change the perspective of the chart, determine whether the chart uses the simple or realistic schemes or your own custom scheme, and the illumination that controls where the shadows will fall.

Rotation and Perspective

To rotate a 3D chart or view it in perspective, enter the required values on the *Perspective* tab of the 3D View dialog. You can also rotate 3D charts interactively; see page 20.

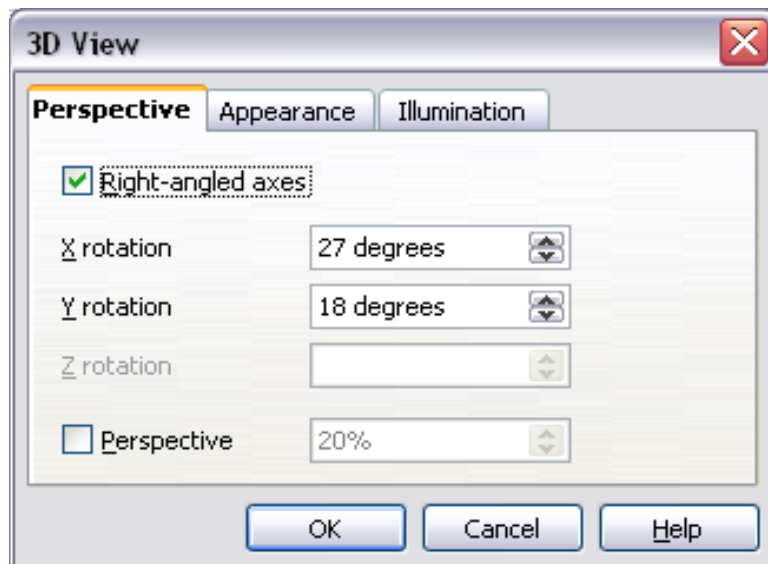


Figure 14: Rotating a chart

Here are some hints for using the *Perspective* tab:

- Set all angles to 0 for a front view of the chart. Pie charts and donut charts are shown as circles.
- With Right-angled axes enabled, you can rotate the chart contents only in the X and Y directions; parallel to the chart borders.
- An x value of 90, with y and z set to 0, provides a view from the top of the chart. With x set to -90, the view is from the bottom of the chart.
- The rotations are applied in the following order: x first, then y, and z last.
- When shading is enabled, and you rotate a chart, the lights are rotated as if they are fixed to the chart.
- The rotation axes always relate to the page, not to the chart's axes. This is different from some other chart programs.
- Select the *Perspective* option to view the chart in central perspective, like through a camera lens, instead of using a parallel projection.
- Set the focus length in the box next to the *Perspective* option (it becomes active when you select the option). 100% gives a perspective view where a far edge in the chart looks approximately half as big as a near edge.

Appearance

Use the *Appearance* tab to modify some aspects of a 3D chart's appearance.

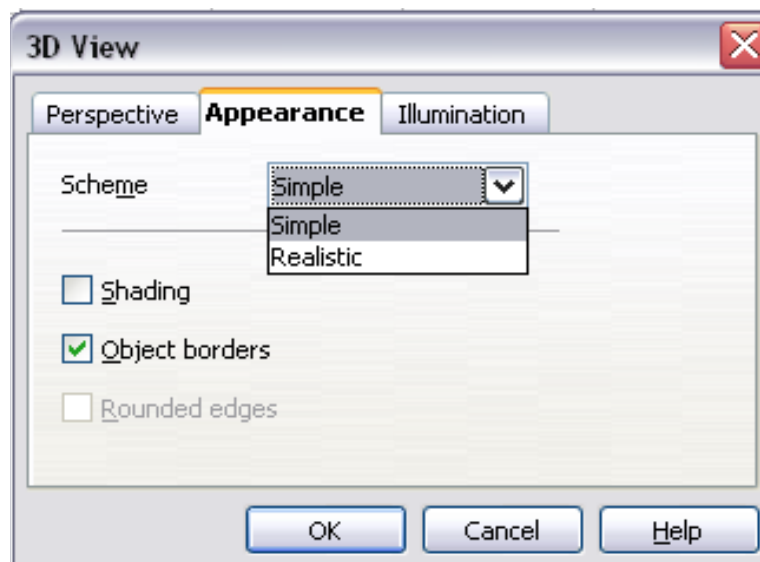


Figure 15: Modifying appearance of 3D chart

Select a scheme from the list box. When you select a scheme, the options and the light sources are set accordingly. If you select or deselect a combination of options that are not given by the Realistic or Simple schemes, you can create a Custom scheme.

Select **Shading** to use the Gouraud method, which applies gradients for a smoother, more realistic look. Otherwise, a flat method is used. The flat method sets a single color and brightness for each polygon. The edges are visible, soft gradients and spotlights are not possible. Refer to the *Draw Guide* for more details on shading.

Select **Object Borders** to draw lines along the edges.

Select **Rounded Edges** to smooth the edges of box shapes. In some cases, this option is not available.

Illumination

Use the *Illumination* tab (Figure 16) to set the light sources for the 3D view. Refer to the *Draw Guide* for more details on setting the illumination.

Click any of the eight buttons to switch a directed light source on or off. By default, the second light source is switched on. It is the first of seven normal, uniform light sources. The first light source projects a specular light with highlights.

For the selected light source, you can then choose a color and intensity in the list just below the eight buttons. The brightness values of all lights are added, so use dark colors when you enable multiple lights.

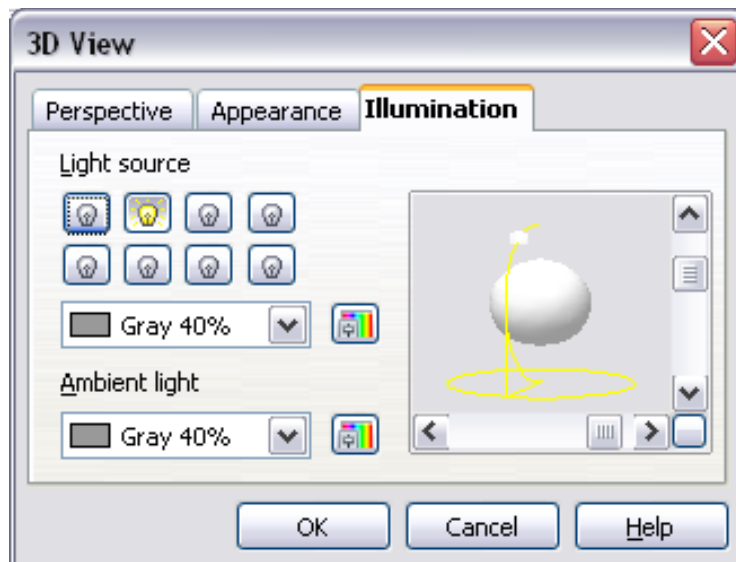


Figure 16: Setting the illumination

Initially, each light source always points at the middle of the object. To change the position of the light source, use the small preview inside this tab. It has two sliders to set the selected light source's vertical and horizontal position.

The button in the corner of the small preview switches the internal illumination model between a sphere and a cube.

Use the *Ambient* light list to specify the nature of light that will shine with uniform intensity from all directions.

Rotating 3D Charts Interactively

In addition to using the Perspective tab of the 3D View dialog to rotate 3D charts, you can also rotate them interactively.

Select the Chart Wall, then hover the mouse pointer over a corner handle or the rotation symbol located in the middle of the chart's information grid. The cursor changes to a rotation icon.

Press and hold the left mouse button and drag the corner in your desired direction. A dashed outline of the chart appears while you drag to help you visualize the result.

Formatting the Chart Elements

Depending on the purpose of your document, for example, a screen presentation or a printed document for a black-and-white publication, you might wish to use more detailed control over the different chart elements to give you what you need.

To format an element, double-click on the chart to put it in edit mode and then left-click on the element you wish to change, for example, one of the axes. The element will be highlighted with green squares. Then, right-click and choose an item from the context menu. Each chart element has its own selection of items. In the following few sections, we explore some of those options.

Formatting Axes and Inserting Grids

Sometimes you need to have a special scale for one of the axes of your chart, or you need smaller grid intervals, or you want to change the formatting of the labels on the

axes. After highlighting the axis you wish to change, right-click and choose one of the items from the pop-up menu.

Choosing **Format > Axis > X Axis** or **Format > Axis > Y Axis** opens the dialog shown in Figure 17, which in this case is the *Scale* tab. The fields available on this dialog depend on the chart type and whether it is 2D or 3D.

On the *Scale* tab, you can choose a logarithmic or linear scale (default), how many marks you need on the line, where the marks are to appear, and the increments (intervals) of the scale. You must first deselect the **Automatic** option to modify the value for any scale. Selecting the **Reverse direction** radial creates a backwards representation of the chart.

On the *Label* tab (Figure 18), you can choose whether to show or hide the labels and how to handle them when they won't all fit neatly into one row (for example, if the words are too long).

Not shown here are the tabs with options for choosing a font and specific font effects, formatting the lines, positioning the elements of the line and interval marks, and selecting effects relating to Asian Typography if this function has been activated in **Tools > Options > Languages**.

X Axis

Scale Positioning Line Label Numbers Font Font Effects

Scale

☐ Reverse direction

☐ Logarithmic scale

Minimum 6 ☒ Automatic

Maximum 18 ☒ Automatic

Major interval 2 ☒ Automatic

Minor interval count 2 ☒ Automatic

Figure 17: Formatting axis scales

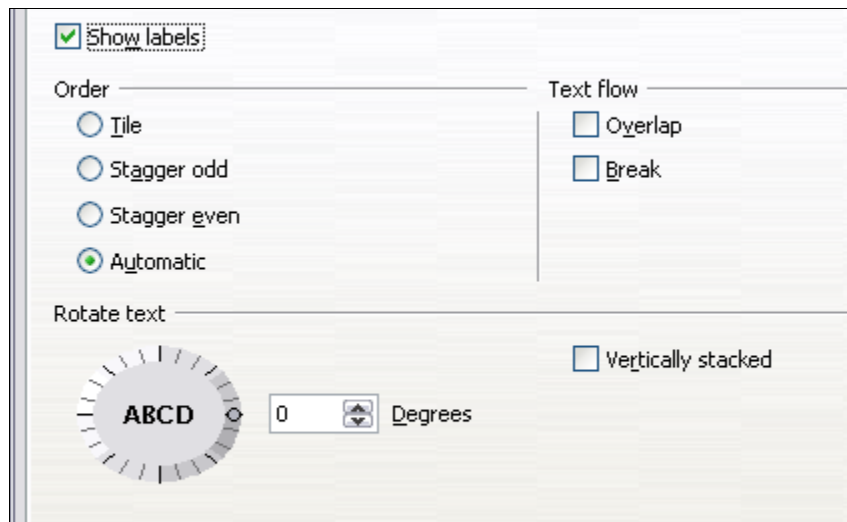


Figure 18: Formatting axis labels

Formatting Data Labels

You can choose properties for the labels of the data series. The labels are inserted with the menu **Insert > Data Labels**. You can adjust the labels later by right-clicking on the data series and choosing the command **Format Data Labels**. This opens a dialog with several tabs where you can change the color of the label text, the size of the font, and other attributes. The *Data Label* tab is shown in Figure 10.

On the *Data Label* tab, you can choose whether to:

- Show the labels as text
- Show numeric values as a percentage or a number
- Include the legend box as part of the label

The text for labels is taken from column labels; it cannot be changed there. If such text needs to be abbreviated, or if it did not label your graph as expected, you must change it in the original data table.

Hierarchical Axis Labels

The axis labels can be derived from two columns to make a hierarchical label structure. Figure 19 shows how to set up the *Data Series* step of the Chart Wizard to make an x-axis with two levels of labels. Notice that the cell range in the *Categories* box spans columns A and B.

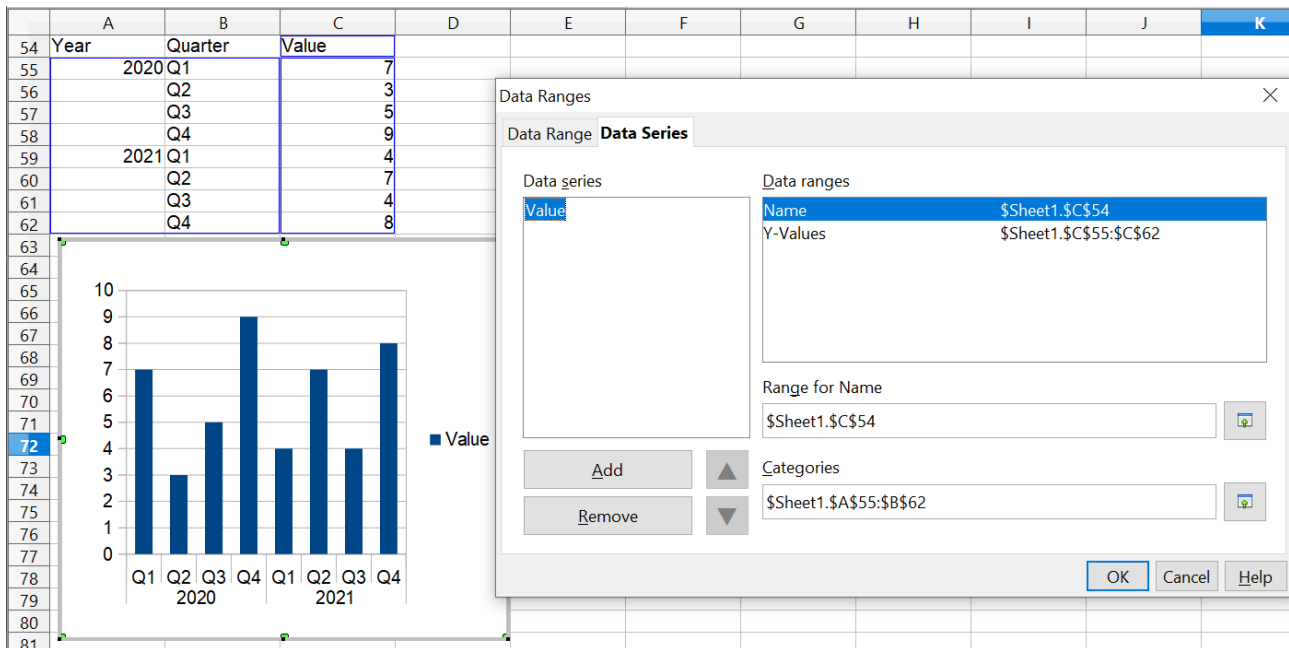


Figure 19: Setting up Hierarchical x-axis labels

Choosing and Formatting Symbols

In line and scatter charts, the symbols representing the points can be changed to a different symbol shape or color through the object properties dialog. Select the data series you wish to change, right-click, and choose **Format Data Series** from the context menu. You can reach this dialog by double-clicking the area of data you would like to change.

On the *Line* tab of the Data Series dialog, in the Icon section, choose from the drop-down list **Select > Symbols**. Here you can choose no symbol, a symbol from a built-in selection, or a more visually interesting range from the gallery. If you have pictures you need to use instead, you can insert them using **Select > From file**.

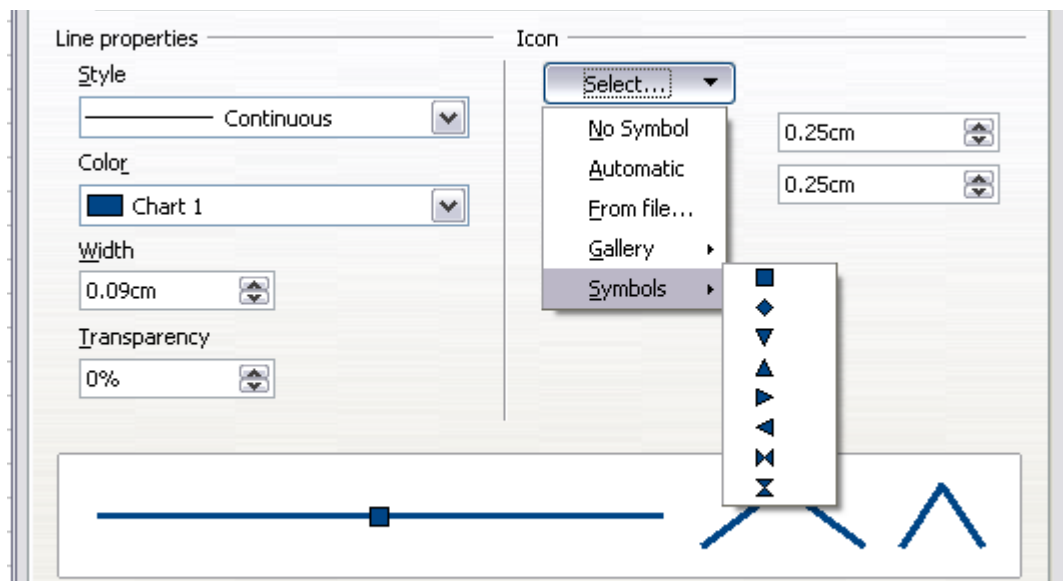
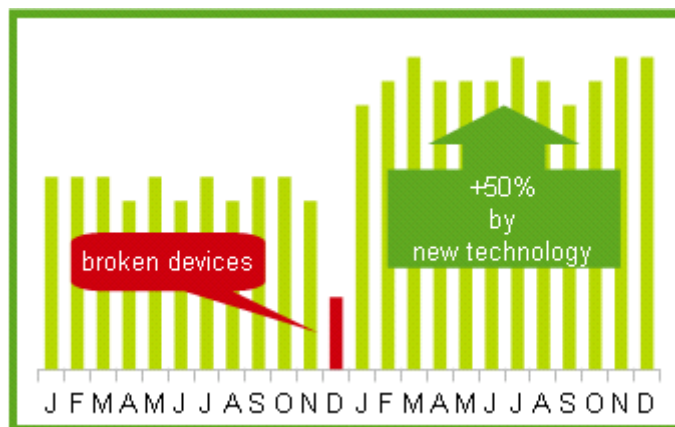


Figure 20: Symbol selection

Adding Drawing Objects to Charts

As in other OpenOffice components, you can use the drawing toolbar to add simple shapes such as lines, rectangles, text objects, or more complex shapes such as symbols or block arrows. This toolbar is located at the bottom of the screen and automatically appears when a chart is generated. Use these additional shapes to add explanatory notes to your chart, as demonstrated in the figure below.



To format the drawing objects, right-click and choose your changes from the context menu.

Resizing and Moving the Chart

You can resize or move all elements of a chart simultaneously, in two ways: interactively or by using the Position and Size dialog. You may use a combination of these methods: interactive for quick and easy changes, and the dialog for precise sizing and positioning.

To resize a chart interactively:

- 1) Click once on the chart to select it. Green sizing handles appear around the chart.
- 2) To increase or decrease the size of the chart, click and drag one of the markers in one of the four corners of the chart. To maintain the correct ratio of the sides, hold the *Shift* key down while you click and drag.

To move a chart interactively:

- 1) Click on the chart to select it. Green sizing handles appear around the chart.
- 2) Hover the mouse pointer anywhere over the chart. When it changes to the move icon, click and drag the chart to its new location.
- 3) Release the mouse button when the element is in the desired position.

Using the Position and Size Dialog

To resize or move a chart using the Position and Size dialog:

- 1) Click on the chart to select it. Green sizing handles appear around the chart.
- 2) Right-click and choose **Position and Size** from the pop-up menu.
- 3) Make your choices on this dialog.

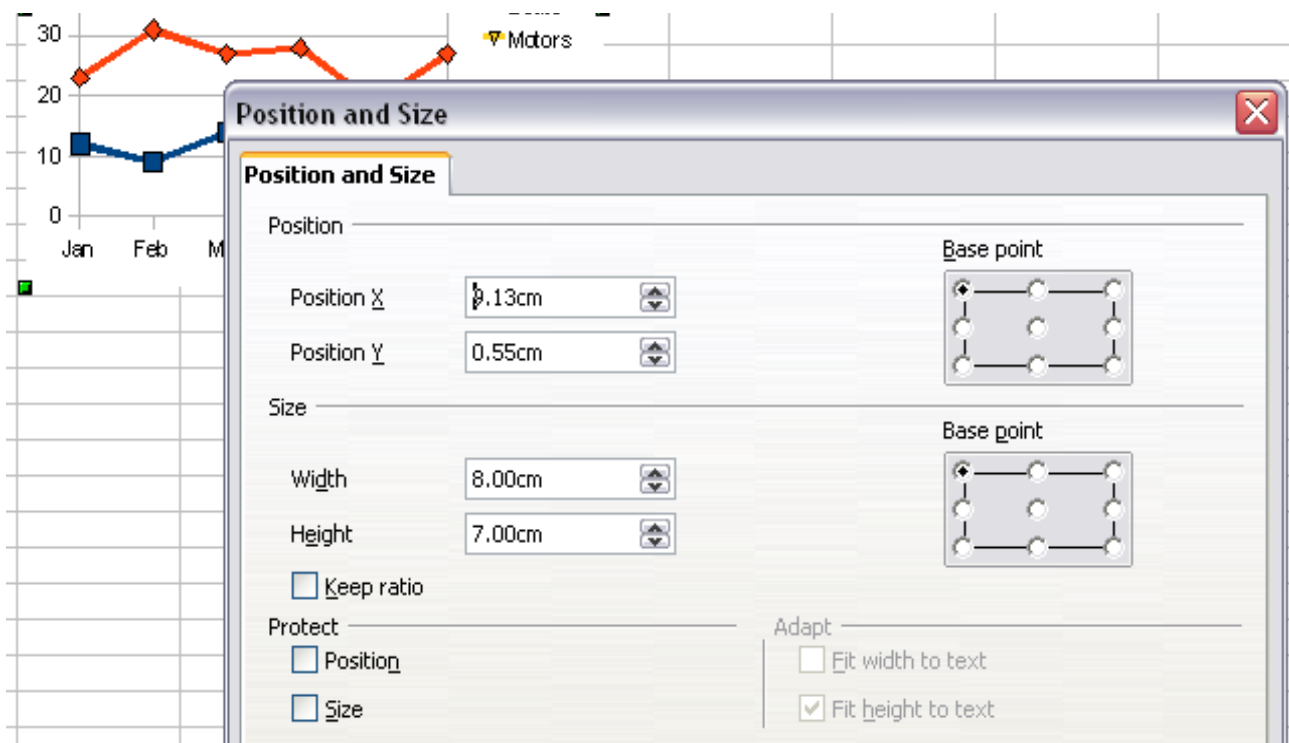


Figure 21: Defining the position and size of an object

Position is defined as an X, Y coordinate relative to a fixed point (the *base point*), typically located at the upper left of the document. You can temporarily change this base point to simplify positioning or dimensioning (click on the spot corresponding to the location of the base point in either of the two selection windows on the right side of the dialog—upper for positioning or lower for dimensioning).

The possible base point positions correspond to the handles on the selection frame plus a central point. The change in position lasts only as long as the dialog is open; when you close it, Calc resets the base point to the standard position.

Tip The **Keep ratio** option is very useful. Select it to keep the ratio of width to height fixed while you change the size of an object.

Size and position can be protected so they can't be accidentally changed. Select the appropriate options.

Tip If you cannot move an object, check to see if its position is protected.

Gallery of Chart Types

It's important to remember that while your data can be presented with many different charts, the message you want to convey to your audience dictates the chart you ultimately use. The following sections present examples of the types of charts that Calc provides, with some of the tweaks each sort can have and some notes as to what purpose you might have for that chart type. For further details, see the OpenOffice Help index.

Column Charts

Column charts are commonly used for data that come from categories. You might plot sales (numeric) for different companies (categorical). They are also widely used to show data over time if the emphasis is on aggregating data during periods of time. An example of this would be showing sales per month. They are best for charts that have a relatively small number of data points. (For a larger time series, a line chart would be better.) It is the default chart type, as it is one of the most useful and easiest to understand.

Bar Charts

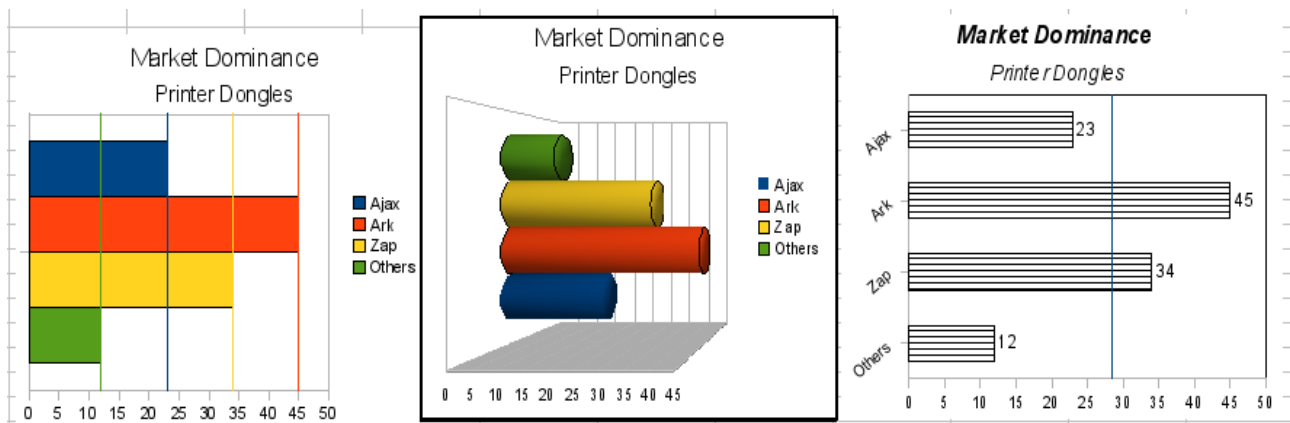


Figure 22: Three bar graph treatments.

Bar charts are column charts with categories on the y-axis. They are excellent for giving an immediate visual impact to data comparisons, especially when time is not an important factor, such as comparing the popularity of a few products in a marketplace.

- The first chart in Figure 22 is achieved quite simply by using the chart wizard with **Insert > Grids**, deselecting y-axis, and using **Insert > Mean Value Lines**.
- The second chart in the figure is the 3D option in the chart wizard with a simple border and the 3D chart area twisted around.
- The third chart in the figure has the legend removed, and the labels show the names of the companies on the axis instead. We also changed the colors to a hatch pattern.

Pie Charts

Pie charts are commonly used to compare proportions. For example, a pie chart would be appropriate if you needed to compare departmental spending, such as what a department spent on different items or what different departments spent. These charts work best with smaller numbers of values; no more than about half a dozen should be included because using more can cause the visual impact to fade. At the same time, it's difficult to compare precisely the size of the slices in a pie chart. As a result, this chart type should not be used if small differences are important.

The Chart Wizard guesses the series you want to include in your pie chart. That being the case, you might need to adjust the series initially, on the Wizard's Data Ranges page, if you know you want a pie chart. This adjustment can also be made by using the **Format > Data Ranges > Data Series** dialog.

You can do very interesting things with a pie chart, especially if you make it into a 3D chart. After that transformation, it can be tilted, given shadows, and come to resemble a work of art.

You can choose an option in the Chart Wizard to explode the pie chart, but this is an all-or-nothing option. To accentuate one piece of the pie, you can separate out one piece by carefully highlighting it after you have finished with the Chart Wizard and dragging it out of the group. To highlight a piece, click once on its edge to select the whole data series and then again to select the individual piece. The two clicks must be well separated in time. When you reposition the piece, you might need to enlarge the chart area again to regain the original size of the pieces.

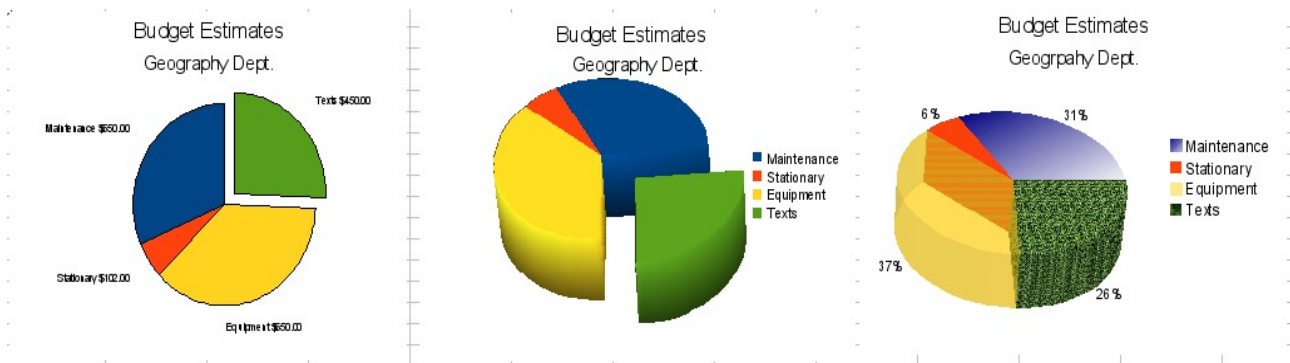


Figure 23: Pie charts

The effects achieved in Figure 23 are explained below.

- The first example is a 2D pie chart with one part of the pie exploded. To produce this type of chart, choose **Insert > Legend** and deselect **Display legend**. Choose **Insert > Data Labels** and choose **Show value as number**. Then, carefully select the piece you wish to highlight, move the cursor to the edge of the piece, click (the piece will have nine green highlight squares to mark it), and then drag it out from the rest of the pieces. The pieces will decrease in size, so you need to highlight the chart wall and drag it to a corner to increase the size.
- The second example is a 3D pie chart with realistic schema and illumination. With a completed 3D pie chart, choose **Format > 3D view > Illumination** where you can change the direction of the light, the color of the ambient light, and the depth of the shade. We also adjusted the 3D angle of the disc in the *Perspective* dialog on the same set of tabs. The chart updates as you make changes, so you can immediately see the effects.
If you want to separate one of the pieces, click on it carefully; you should see a wireframe highlight. Drag it out with the mouse and then, if necessary, increase the size of the chart wall.
- The third example is a 3D pie chart with different fill effects in each portion of the pie. Choose **Insert > Data labels** and select **Show value as percentage**. Carefully select each piece so that it has a wireframe, then highlight and right-click to get the object properties dialog. Choose the *Area* tab. For one of the pieces, we chose a bitmap effect; for another, we selected a gradient feature; for the third, we used the *Transparency* tab and adjusted the transparency to 50%.

Donut Charts

Donut charts are a variation of the pie chart. To create one, choose Pie in the Chart Type dialog, and choose the third or fourth type of pie chart. For more variety, consider selecting **3D Look**.

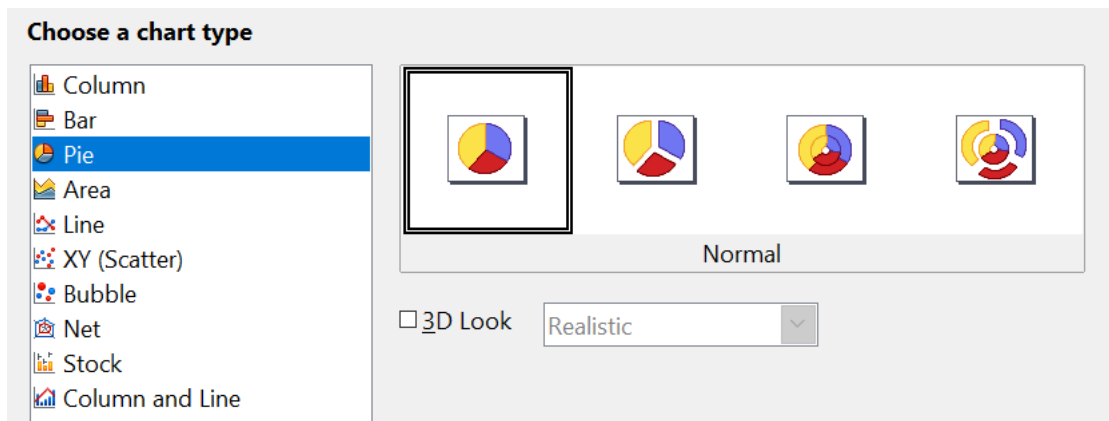


Figure 24: Choosing a donut chart

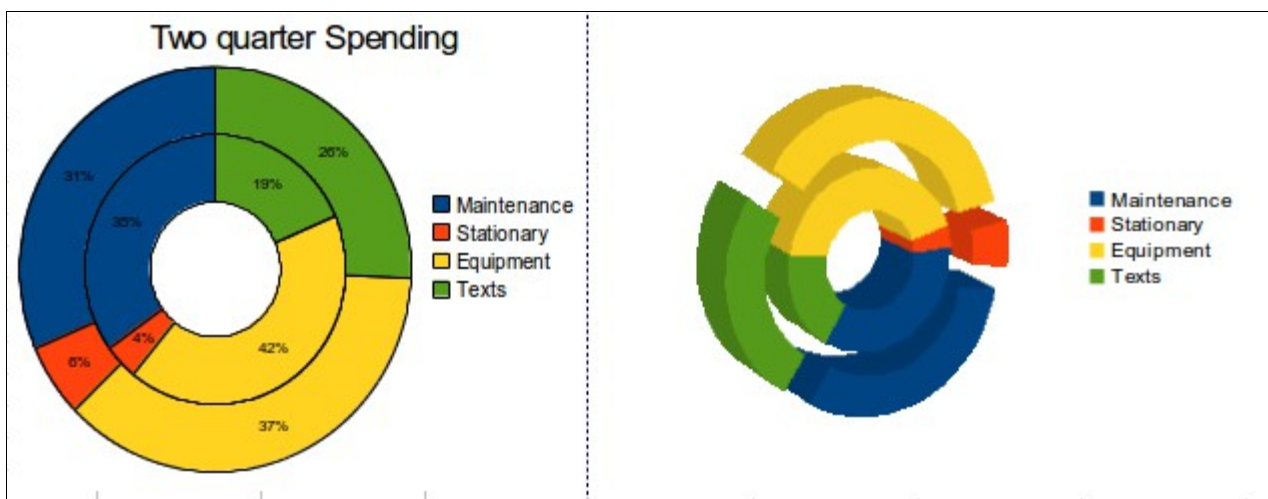


Figure 25: Examples of donut charts

Area Charts

An *area chart* is a version of a line or column graph. It may be useful where you wish to emphasize volume of change. Area charts have a greater visual impact than line charts, but the data you use will make a difference.

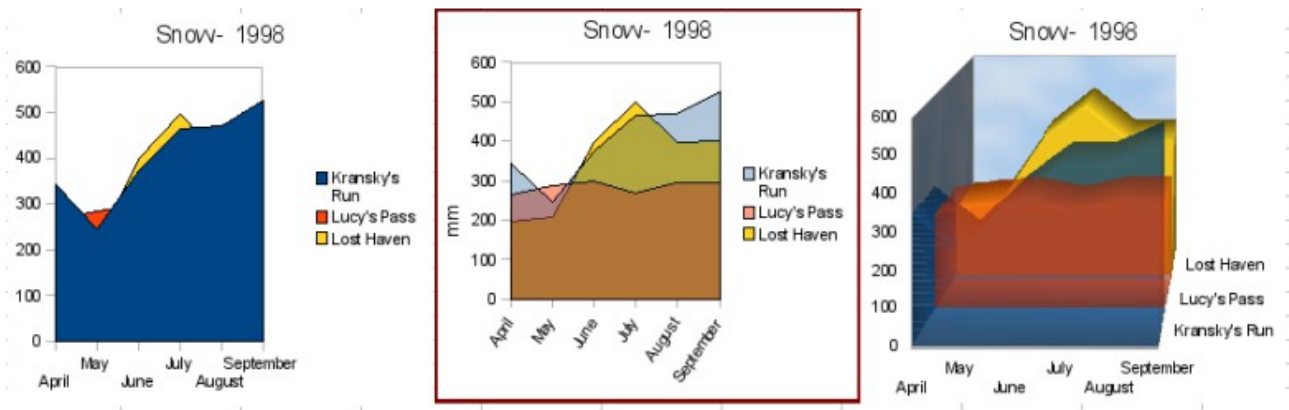


Figure 26: Area charts—the good, the bad, and the ugly

As shown in Figure 26, an area chart is sometimes tricky to use. This may be one good reason to use transparency values in an area chart. After setting up the basic chart using the Chart Wizard, do the following:

- Right-click on the y-axis and choose **Delete Major Grid**. As the data overlaps, some of it is lost behind the first data series. This is not what you want. A better solution is shown in the highlighted chart and is explained in the next point.
- After deselecting the y-axis grid, right-click on each data series in turn and choose **Format Data Series**. On the *Transparency* tab, set Transparency to 50%. The transparency makes it easy to see the data hidden behind the first data series. Now, right-click on the x-axis and choose **Format Axis**. On the *Label* tab, choose **Tile** in the *Order* section and set the Text orientation to 55 degrees. This places the long labels at an angle.
- To create the third variation, after doing the steps above, right-click and choose **Chart Type**. Choose the **3D Look** option and select **Realistic** from the drop-down list. We also twisted the chart area around and gave the chart wall a picture of the sky. As you can see, the legend turns into labels on the z-axis. Overall, though it is visually more appealing, it is more difficult to see the point you are trying to make with the data.

Other ways of visualizing the same data series are represented by the stacked area chart or the percentage stacked area chart.

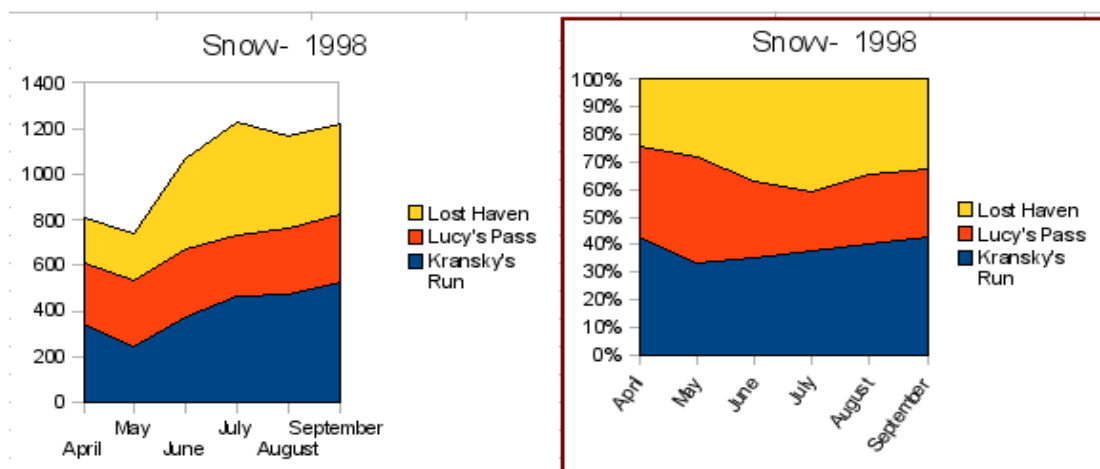


Figure 27: Stacked and percentage stacked area charts

The first does what it says: each number of each series is added to the others so that it shows an overall volume but not a comparison of the data. The percentage stacked chart shows each value in the series as a part of the whole. For example, in June, all three values are added together, and that number represents 100%. The individual values are a percentage of that. Many charts have varieties with this option.

Line Charts

A *line chart* shows a series of data but with categories rather than numeric values on the x-axis. Line charts are commonly used when time is marked with names of months or quarters or a year. As such, this chart type is ideal for raw data. It's also very effective for charts with lots of data that show trends over time when what you want is to emphasize continuity. It's important to remember that the x-axis shows categories and spaces them equally. If you had month labels of April, May, and July, the axis spacing from April to May would be the same as from May to July. This can produce deceptive interpretations.

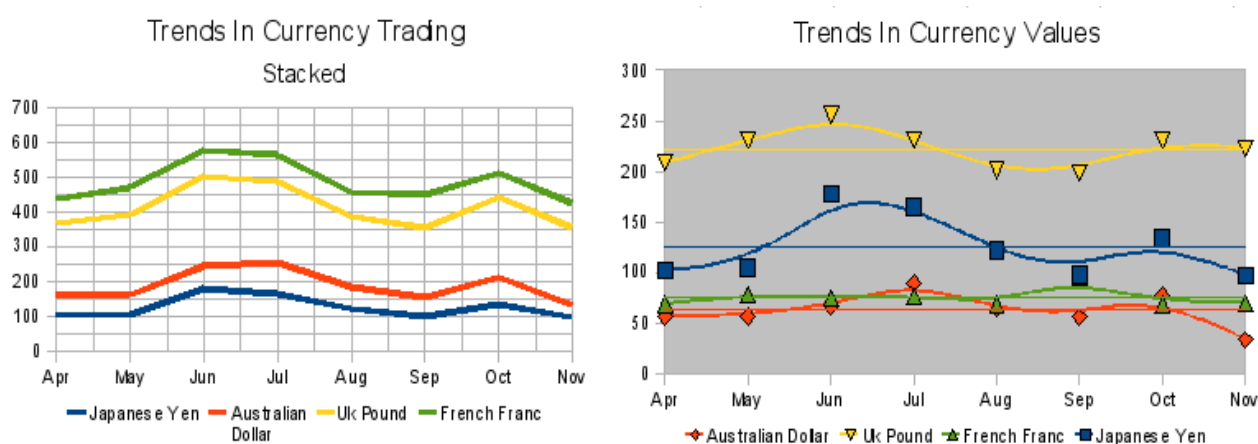


Figure 28: Line charts

Scatter or XY Charts

Scatter charts are great for visualizing data when both axes represent numeric values. A chemical reaction rate measured at different temperatures or income at different ages are examples of such data. The trend line equation inserted with **Insert > Trend Lines** can be used for quantitative predictions, and the data placement correctly accounts for any irregularity in the values of data points. This type of chart is the best for displaying accurate time series, but the date or time values used must be entered as numeric values. This can be checked with **View > Value Highlighting**, which will show values in blue if they are numeric.

Scatter charts may surprise those unfamiliar with how they work. While constructing the chart, if you choose **Data Range > Data series in rows**, the first row of data represents the x-axis. The rest of the rows of data are then compared against the first row data. Figure 29 shows a comparison of three currencies with the Japanese Yen. Even though the table presents the monthly series, the chart does not. In fact, the Japanese Yen does not appear in the legend; it is merely used as the x-axis value that all the other data series are compared against.

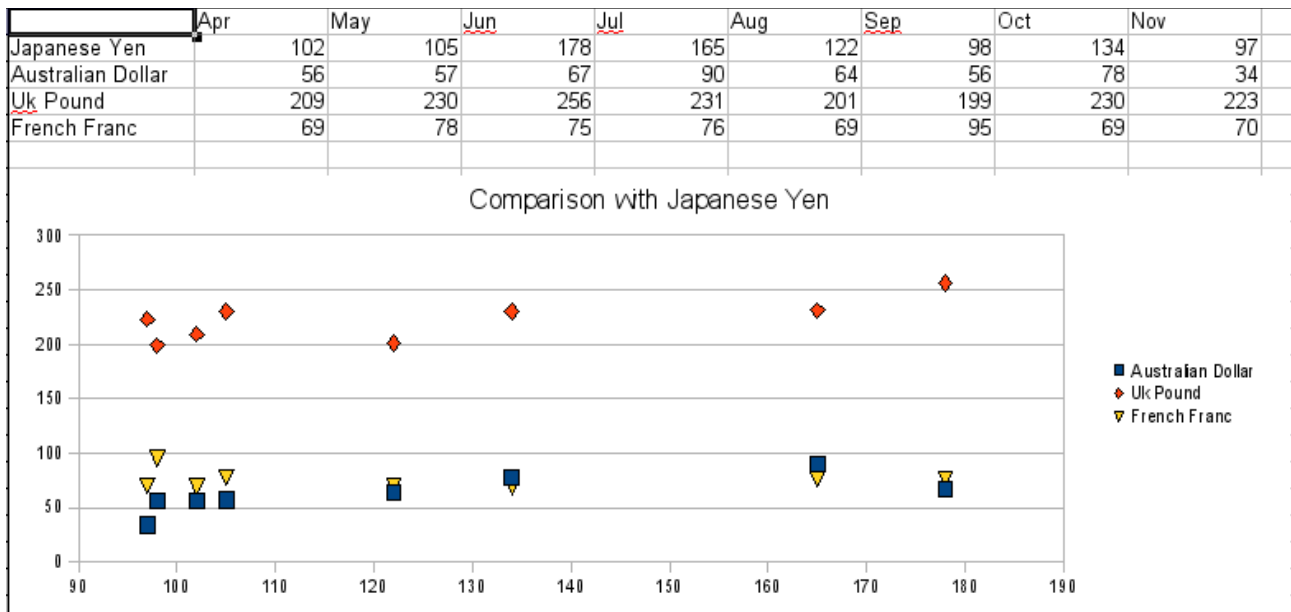


Figure 29: A particularly volatile time in the world currency market.

Bubble Charts

A *bubble chart* is a variation of a scatter chart in which the data points are replaced with bubbles. It shows the relations of three variables in two dimensions. Two variables are used for the position on the x-axis and y-axis, while the third is shown as the relative size of each bubble. One or more data series can be included in a single chart.

Bubble charts are often used to present financial data. The data series dialog for a bubble chart has an entry to define the data range for the bubbles and their sizes.

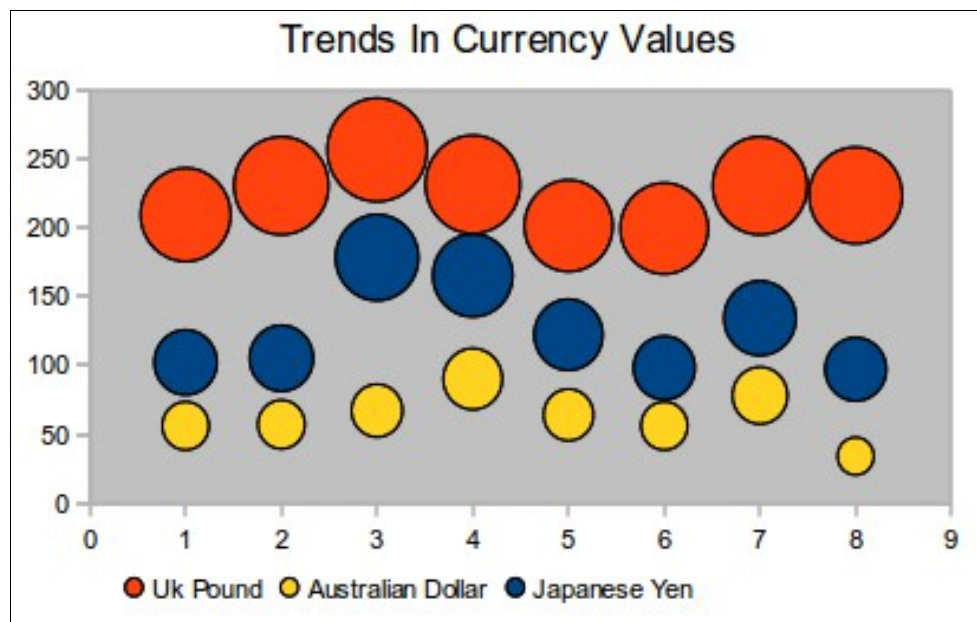


Figure 30: Bubble chart showing three data series

Net Charts

A *net chart* is similar to a polar or radar chart. It is useful for comparing data that is not a time series but shows different circumstances, such as variables in a scientific experiment or direction. The poles of the net chart are equivalent to the y-axis of other charts. Generally, between three and eight axes are best; any more and this type of chart becomes confusing. Before and after values can be plotted on the same chart, or perhaps expected and real results, to compare differences.

Figure 31 shows two types of net charts:

- (Left): A plain net chart without grids or lines, only points.
- (Right): A net chart with lines, points, and a grid. Axis colors and labels have been changed. The chart area color has been changed with a gradient effect. The points on this graph have been changed to fancy 3D spheres.

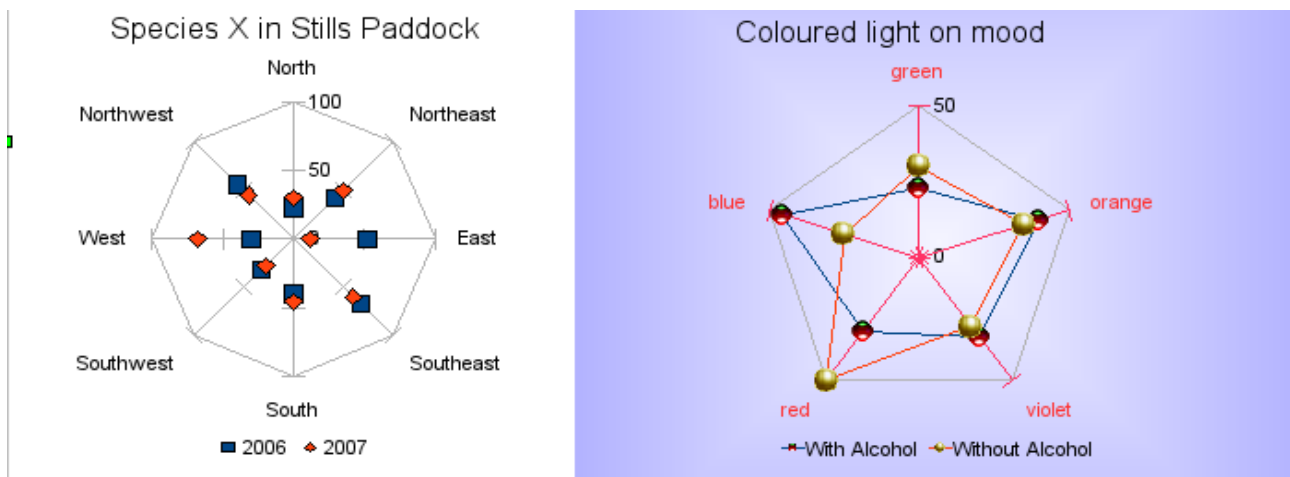


Figure 31: Two net diagrams showing fabricated data from totally fictional experiments.

Other varieties of net charts can be made to show the data series as stacked numbers or stacked percentages. The series can also be filled with a color (Figure 32). Partial transparency is often best for showing all the series at once.

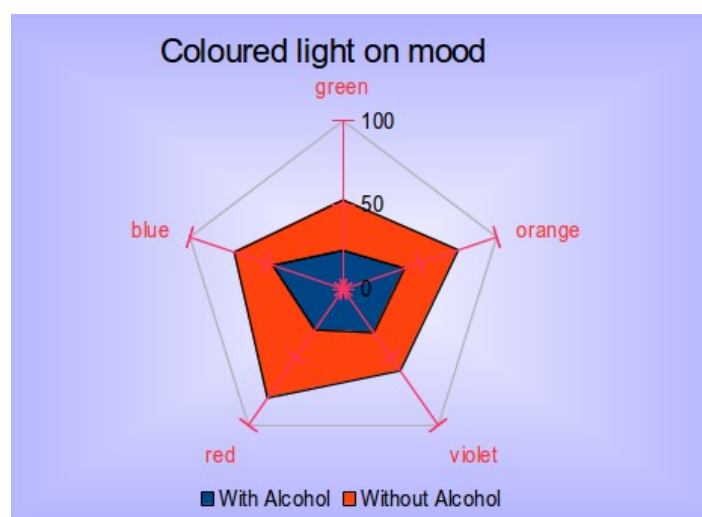


Figure 32: Filled net or radar chart

Stock Charts

A *stock chart* is a specialized column graph specifically for stocks and shares. You can choose traditional lines, candlestick, and two-column type charts. The data required for these charts is quite specialized, with series for opening price, closing price, and high and low prices. In this type of chart, the x-axis represents a time series.

When you set up a stock chart in the Chart Wizard, the Data Series dialog is very important. You need to tell it which series represents the opening price, the closing price, the high and low price of the stock, and so on.

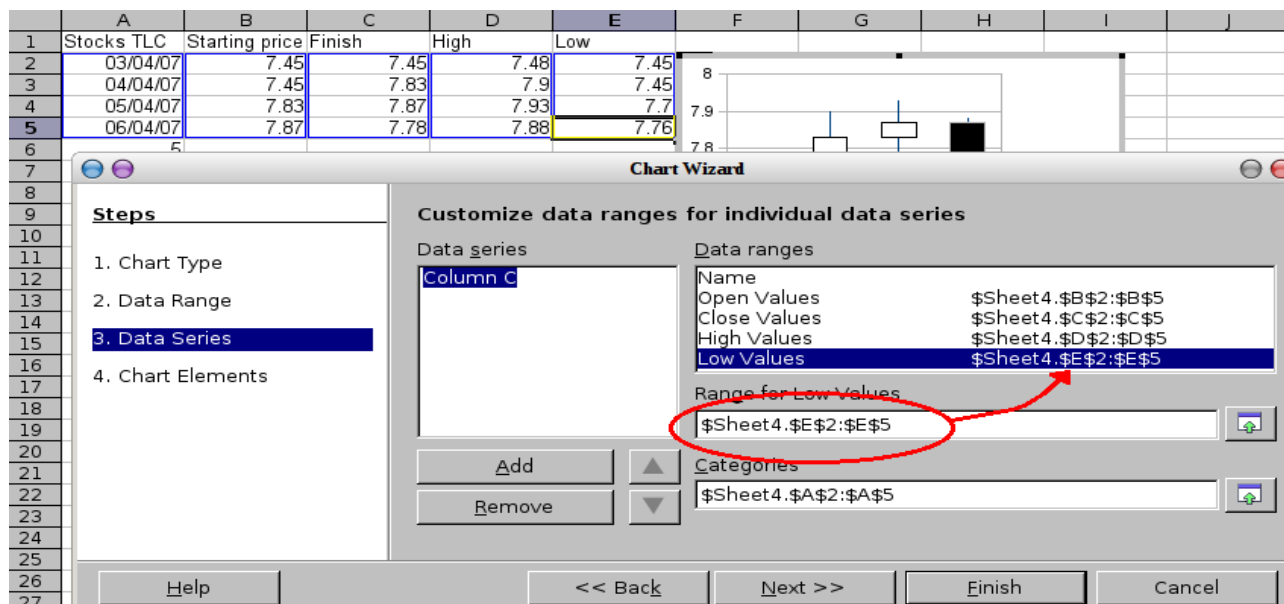


Figure 33: Adjusting data series for stock charts.

A nice touch is that OpenOffice stock charts color-code the rising and falling shares: white for rising, black for falling in the candlestick chart, and red and blue in the traditional line chart.

Column and Line Charts

A *column and line chart* is a combination of two other chart types. It is useful for combining two distinct but related data series like sales over time (column) with a profit margin trends (line).

You can choose the number of columns and lines in the Chart Wizard. For example, you might have two columns with two lines to represent two product lines with the sales figures and profit margins of both.

The chart in Figure 34 shows manufacturing cost and profit data for two products over a period of time (six months in 2007). To create this chart, highlight the table and start the Chart Wizard. Choose the Column and Line chart type with two lines and the data series in rows. Then, give it a title to highlight the aspect you want to show. The lines are different colors at this stage and don't reflect the product relationships. When you finish with the Chart Wizard, highlight the chart, click on the line, right-click and choose **Format Data Series**.

On this tab, there are a few things to change: The colors should match the products so both Ark Manufacturing and its profit line are blue and Prall Manufacturing is red. The lines need to be more noticeable, so make them thicker by increasing the width to 0.08.

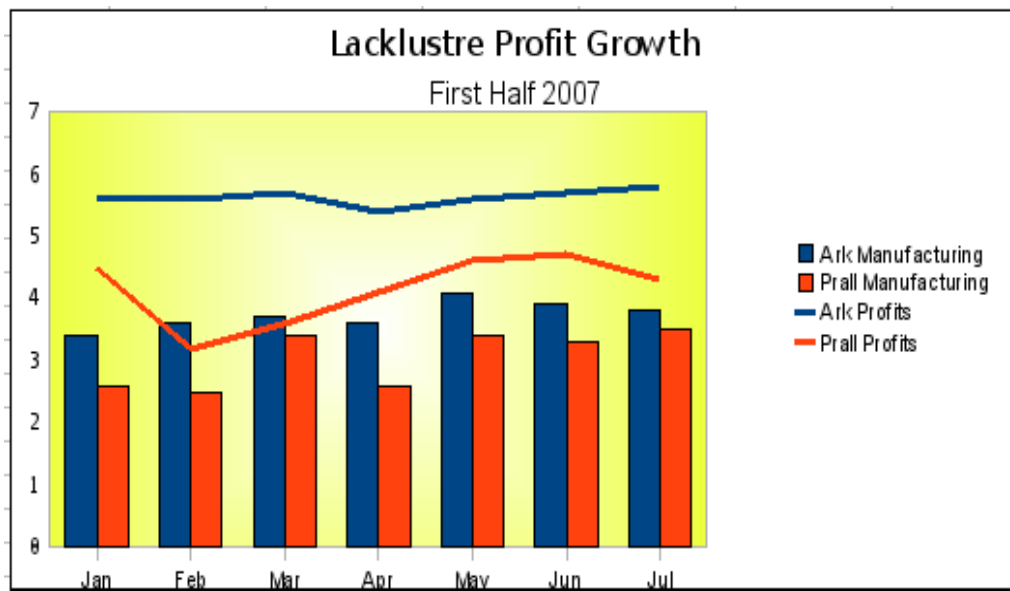


Figure 34: Column and line chart

For the background, highlight the chart wall, right-click it, and choose **Format Wall**. On the *Area* tab, change the drop-down box to show Gradient. Choose one of the preset gradient patterns and make it lighter by going to the *Transparency* tab and making the gradient 50% transparent.

To make the chart look cleaner without the grid, go to **Insert > Grids** and deselect the X-axis option.